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PLANT DOCTOR

**Early Detection
Strong Protection**



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Problem Statement

There are many challenges that farmers face in protecting crops, such as:

Farmers face three main problems:

- Late disease detection damages crops and reduces yield.
- Wrong pesticide use increases costs and harms soil health.
- Lack of expert advice leaves farmers relying on guesswork.

These challenges cause heavy losses and lower farm productivity.



Barriers to Agri-Tech Adoption



Dependence on Informal Sources

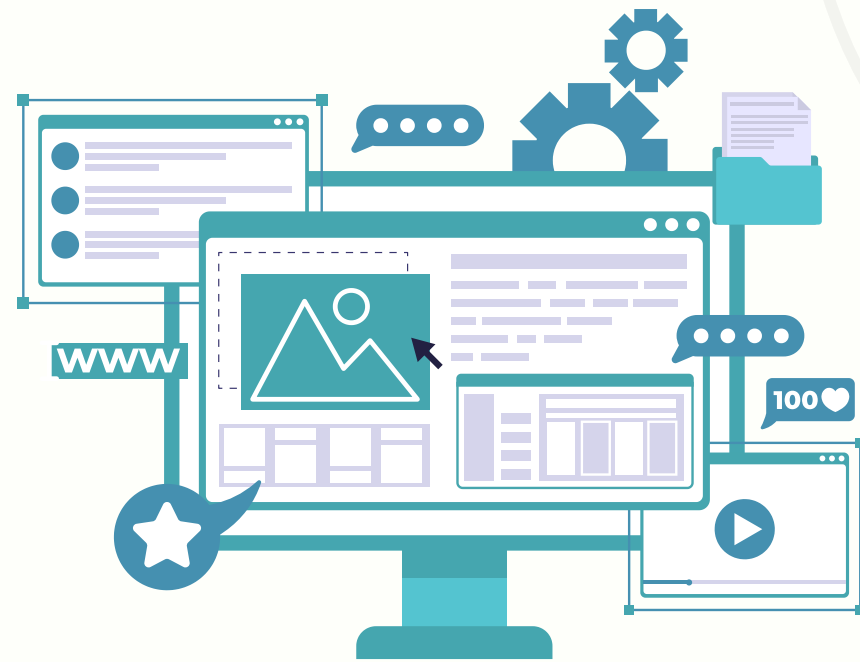


Lack of Personalized Advisory



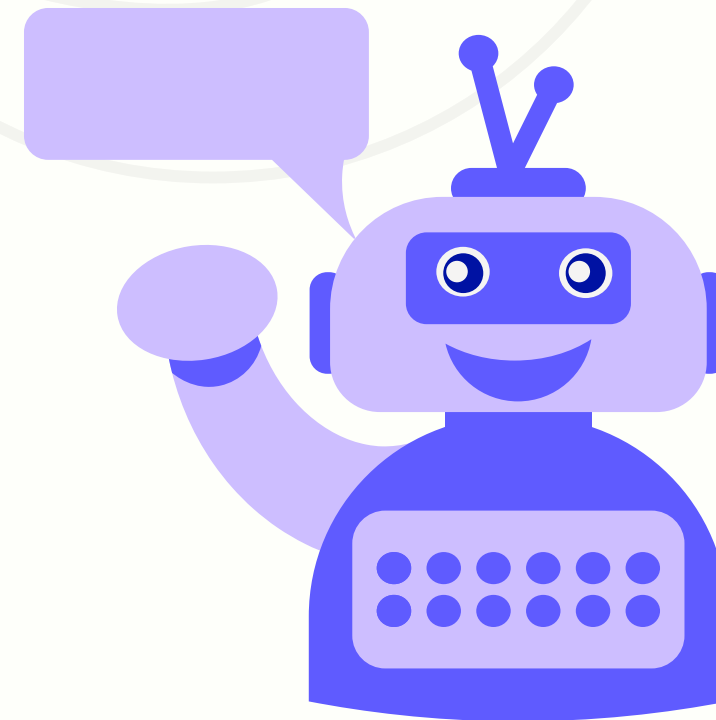
Our Solution

Plant doctor creates a positive impact for farmers and agriculture by:



AI-powered mobile & web app →

Instant crop disease detection using a simple photo.



24/7 Chatbot advisory →

Clear guidance on treatment, prevention, and safe pesticide dosage.



Weather-driven alerts →

Smarter, timely spraying decisions that save cost and protect soil.

Our Solution

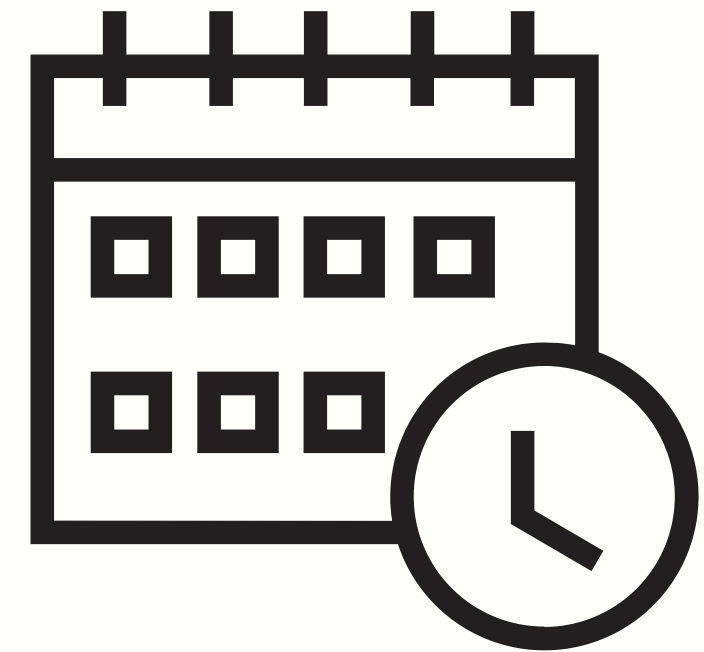


Plant Doctor creates a positive impact for farmers and agriculture by:



Soil Health

Soil health recommendations and fertilizer guidance.

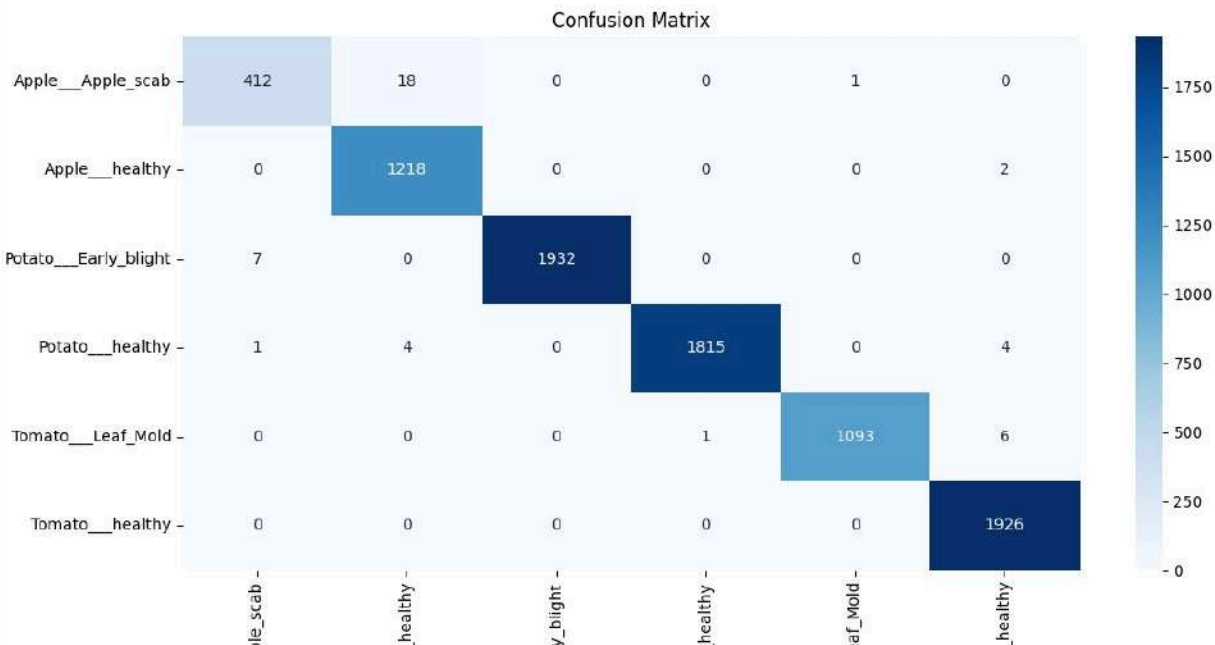
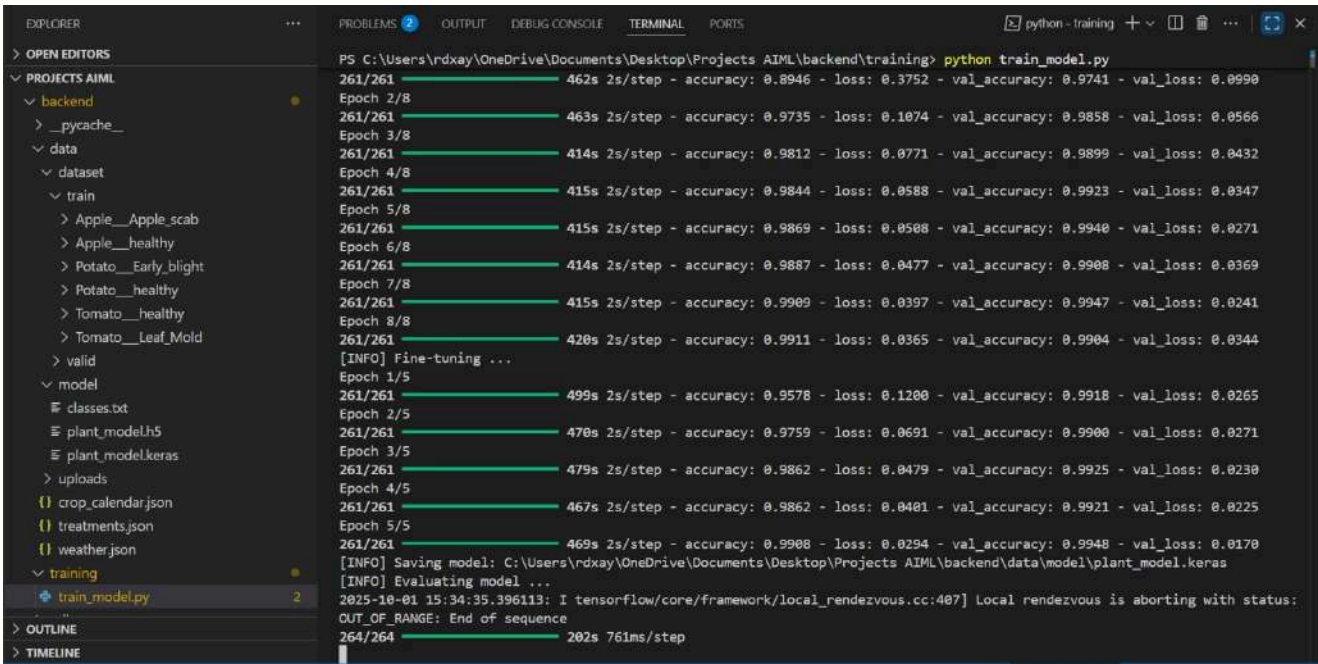


Crop Calendar

A systematic schedule that guides farmers on the right time for sowing, growing, and harvesting crops according to season and climate



Model Training Results



Accuracy (Final Evaluation Results)

Overall Accuracy: 99.5%

Macro Average Precision: 99.3%

Macro Average Recall: 99.0%

Macro Average F1-Score: 99.1%

Weighted Average Accuracy: 99.5%

Class-wise Performance:-

Apple Scab: Precision 98.1% | Recall 95.6% | F1-score 96.8%

Apple Healthy: Precision 98.2% | Recall 99.8% | F1-score 99.0%

Potato Early Blight: Precision 100% | Recall 96.0% | F1-score 98.0%

Potato Healthy: Precision 99.9% | Recall 99.5% | F1-score 99.7%

Tomato Leaf Mold: Precision 99.4% | Recall 99.6% | F1-score 99.5%

Tomato Healthy: Precision 99.4% | Recall 100% | F1-score 99.7%

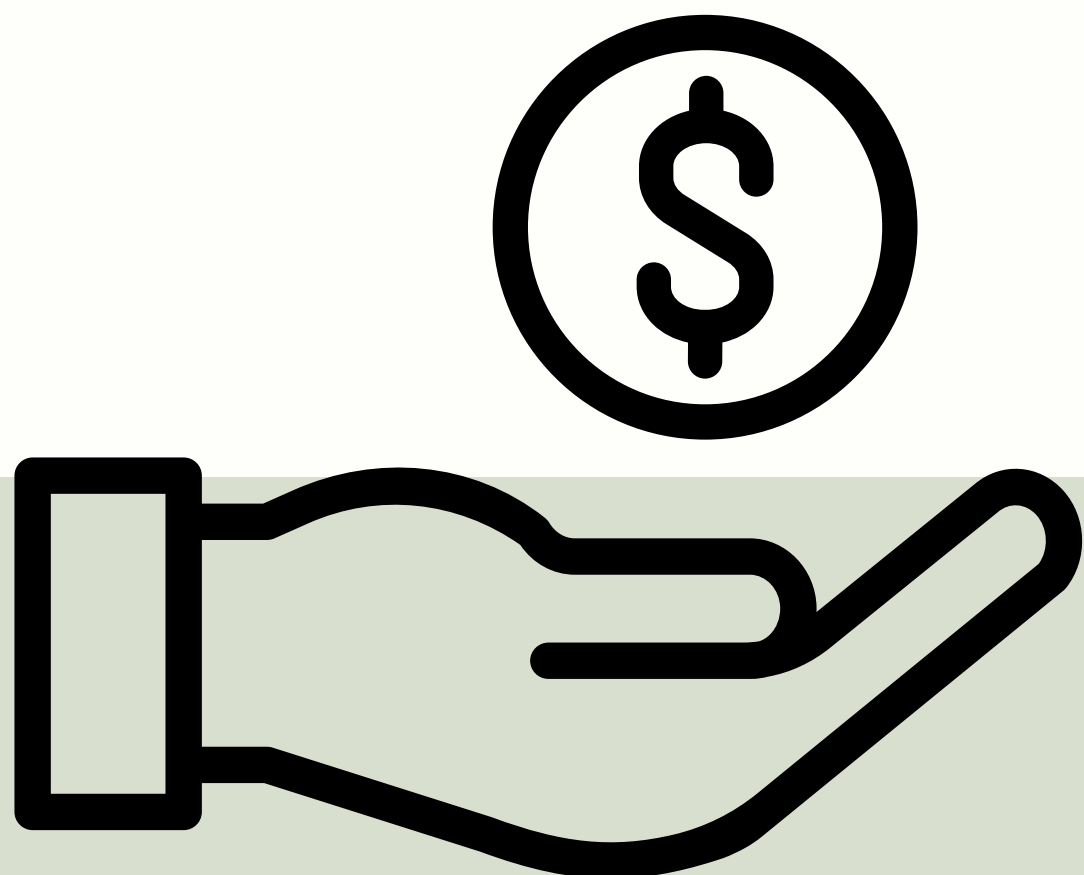


How We Differ

- Works offline (TFLite/ONNX) in low-network villages.
- Integrated dosage calculator → prevents chemical misuse.
- Weather + AI fusion → unique decision support.
- Multilingual chatbot → farmer-friendly & accessible.

Unique Selling Points ✨

- 🌿 Multilingual chatbot → local farmers use easily.
- ⚡ Offline-first → works in villages.
- 🤖 AI + Weather Combo → smarter farm planning.
- ☁ All-in-one portal → crops, soil, medicine info.



जय जवान जय किसान

THANK YOU!

"Smart Farming Today for a Sustainable Tomorrow"



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